

Noncontact Warming Therapy, Ultrasound Therapy, and Fluorescence Imaging for Wounds

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Community Plan Policy

- [Noncontact Warming Therapy, Ultrasound Therapy, and Fluorescence Imaging for Wounds](#)

Application

UnitedHealthcare Commercial

This Medical Policy applies to UnitedHealthcare Commercial benefit plans.

UnitedHealthcare Individual Exchange

This Medical Policy applies to Individual Exchange benefit plans.

Coverage Rationale

Warming therapy or noncontact normothermic wound therapy and low-frequency ultrasound therapy are unproven and not medically necessary for treating wounds due to insufficient evidence of efficacy.

Noncontact real-time fluorescence wound imaging for bacterial presence is unproven and not medically necessary due to insufficient evidence of safety and/or efficacy.

Applicable Codes

The following list(s) of procedure and/or diagnosis codes is provided for reference purposes only and may not be all inclusive. Listing of a code in this policy does not imply that the service described by the code is a covered or non-covered health service. Benefit coverage for health services is determined by the member specific benefit plan document and applicable laws that may require coverage for a specific service. The inclusion of a code does not imply any right to reimbursement or guarantee claim payment. Other policies and guidelines may apply.

CPT Code	Description
0598T	Real-time fluorescence wound imaging with clinical darkness, to identify location of bacterial wound pathogens and measure wound size, per session; first anatomic site (e.g., lower extremity, right leg)
0599T	Real-time fluorescence wound imaging with clinical darkness, to identify location of bacterial wound pathogens and measure wound size, per session; each additional anatomic site (e.g., upper extremity, left leg) (List separately in addition to code for primary procedure)

CPT Code	Description
97610	Low frequency, non-contact, non-thermal ultrasound, including topical application(s), when performed, wound assessment, and instruction(s) for ongoing care, per day

CPT® is a registered trademark of the American Medical Association

HCPSC Code	Description
A6000	Noncontact wound-warming wound cover for use with the noncontact wound- warming device and warming card
E0231	Noncontact wound-warming device (temperature control unit, AC adapter and power cord) for use with warming card and wound cover
E0232	Warming card for use with the noncontact wound warming device and noncontact wound warming-wound cover

Description of Services

Warming therapy or noncontact normothermic wound therapy uses a noncontact wound cover and a warming unit to apply radiant heat and maintain 100% relative humidity in a wound. The intent is to raise the wound temperature to increase blood flow and oxygen to the wound.

Low-frequency or low-energy ultrasound using the MIST Therapy® System has been developed to provide simultaneous cleansing and debridement of wounds. Treatment with this device involves holding an ultrasonic handset 1 cm away from the wound and applying a saline solution to the handset; a saline mist is generated, which is designed to carry low levels of ultrasonic energy into the wound. According to the device manufacturer, this treatment promotes healing of acute, traumatic, and chronic wounds by stimulating cellular activities that contribute to healing and by cleaning the wound surface.

The MIST Therapy System is a noncontact, low-frequency ultrasound debridement device that has been developed to promote healing of chronic wounds by removing yellow slough, tissue exudates, fibrin, and bacteria from the wound surface. The main components of the MIST Therapy System are an ultrasound generator; a handheld ultrasound transducer; and a single-use, disposable applicator with a bottle reservoir for sterile saline. The transducer tip vibrates 40,000 times per second to generate ultrasound waves at 40 kHz that are carried to the wound via a saline mist. In addition to wound cleaning, ultrasonic energy has been proposed as a means of stimulating angiogenesis, growth factor production, and other cellular activities that contribute to wound healing.

MolecuLight i:X® is a handheld fluorescence imaging device for real-time detection of bacteria in wounds; the violet light illumination captures and documents the presence of bacteria in the wound and surrounding areas. The device provides clinicians with information about the fluorescent characteristics of a wound to assist them in making improved diagnostic and treatment decisions.

Clinical Evidence

Warming Therapy

The safety and efficacy of warming therapy or noncontact normothermic wound therapy (NNWT) for the treatment of chronic wounds have not been established in the published medical literature. Limitations of the existing studies include small samples, lack of controls and/or randomization, and short follow-up times.

Haesler (2023) conducted a wound healing and management evidence summary to evaluate the best-available evidence for the use of local warming or heat therapy in the healing of chronic wounds. The summary was based on a systematic literature search using Joanna Briggs Institute methodologies, drawing on evidence published up to April 2023 from major databases, including MEDLINE, Embase, CINAHL (Cumulative Index to Nursing and Allied Health Literature), and the Cochrane Library. The review examined level 1 to 5 evidence related to the use of radiant or conductive heat therapies in pressure injuries, venous leg ulcers, and diabetic foot ulcers (DFUs), with most studies focusing on NNWT devices that deliver radiant heat in a moist wound environment. Across three small, level 1 studies, one level 2 study of pressure injuries, two small studies of venous leg ulcers, and one small randomized controlled trial (RCT) of DFUs, findings were inconsistent and generally at a high risk of bias, with no clear or reliable evidence demonstrating improved rates of complete healing compared with standard wound care. While some studies reported reductions in wound size or pain, with one small DFU trial showing higher healing rates with heated dressings, the overall evidence was insufficient and methodologically weak. The author concluded that the current evidence does not support recommending local warming or heat therapy to promote

healing in chronic wounds. The limitations highlighted include small sample sizes, a high risk of bias, inconsistent outcome reporting, lack of evidence on conductive heat, and limited safety data, particularly regarding risks such as burns, maceration, and impaired sensation in vulnerable populations.

Yue et al. (2018) conducted a systematic review to assess the effects of local warming therapy (LWT) in treating chronic wounds. The inclusion criteria included RCTs that recruited people with a chronic wound(s) (pressure ulcers, venous leg ulcers, arterial ulcers, DFUs, and arterial ulcers) and compared the effects of LWT with those of standard wound care or other wound healing interventions. The primary outcomes included the time to healing, which was assessed using appropriate survival analysis; proportion of people with DFUs undergoing amputation of the lower limb at any level, including single toes; and proportion of wounds with complete healing. The secondary outcomes consisted of change in wound size, with change expressed as absolute change or relative change; healing rate; and quality of life, which was measured by a validated scale. No RCTs comparing the effects of LWT with those of standard wound care or other wound healing interventions among people with a chronic wound(s) were found. It was therefore not possible to undertake a meta-analysis. The authors concluded that this review highlights the lack of robust evidence for the use of LWT in the treatment of chronic wounds. Thus, no definitive conclusions regarding the use of LWT for treating chronic wounds can be drawn from this review.

Thomas et al. (2005) conducted a randomized controlled comparative study in 41 participants with a stage 3 or stage 4 truncal pressure ulcer of $> 1.0 \text{ cm}^2$. The experimental group was randomized to a radiant-heat dressing device, and the control group was randomized to a hydrocolloid dressing, with or without a calcium alginate filler. They were followed up until healed or for 12 weeks. Eight participants (57%) in the experimental group had complete healing of their pressure ulcer compared with seven participants (44%) with complete healing in the control group ($p = 0.46$). The authors concluded that although a 13% difference in healing rate between the two arms of the study was found, this difference was not statistically significant. This study had a small sample size.

Karr (2003) performed a randomized pilot study to evaluate the use of NNWT in the treatment of wounds associated with osteomyelitis. The study consisted of two arms. The control arm ($n = 11$) received standard wound care, which resulted in complete ulcer healing at an average of 127 days. The treatment arm ($n = 5$) received NNWT, which resulted in complete ulcer healing at an average of 59 days or healing that was 54% faster than that in the control arm. The authors concluded that although NNWT is not a direct treatment for osteomyelitis, this new treatment option results in accelerated healing of wounds associated with osteomyelitis. However, this difference did not reach statistical significance, and median wound healing times did not differ between groups.

Ellis et al. (2003) evaluated 33 individuals with full-thickness pressure sores who were randomized to receive standard care or radiant heat therapy with a Warm-Up[®] device. The Warm-Up device eradicated bacteria in six individuals within 2 weeks of starting treatment compared with none in the standard-care group. The significance of this study is limited by a small sample size.

Low-Frequency Ultrasound

Evidence for the use of low-frequency ultrasound to treat wounds consists of studies that lack adequate sample sizes and proper control groups. Further controlled trials that include larger sample sizes are necessary to demonstrate that low-frequency ultrasound is beneficial for health outcomes in individuals with wounds.

Flores-Escobar et al. (2022) conducted a systematic review and meta-analysis of RCTs to investigate the effect of ultrasound-assisted wound (UAW) debridement in individuals with DFUs. All selected studies were evaluated using the Cochrane risk-of-bias tool to assess the risk of bias for RCTs. A total of eight RCTs met the inclusion criteria and included 263 individuals. Concerning the healing rate, comparing UAW vs the control group, a meta-analysis estimated the pooled odds ratio (OR) to be 2.22 (95% CI, 0.96-5.11; $p = 0.06$), favoring UAW debridement, with low heterogeneity ($\chi^2 = 7.47$; $df = 5$; $p = 0.19$; $I^2 = 33\%$). Time to healing was similar in both groups: 14.25 ± 10.10 weeks in the UAW group vs 13.38 ± 1.99 weeks ($p = 0.87$) in the control group. Wound area reduction (WAR) was greater in the UAW debridement group ($74.58\% \pm 19.21\%$) than in the control group ($56.86\% \pm 25.09\%$), although no differences were observed between them ($p = 0.24$). UAW debridement showed higher healing rates, a greater percentage of WAR, and similar healing times compared with placebo (sham device) and standard of care (SOC) in individuals with DFUs, although no statistical differences were observed between groups. The main limitation of this systematic review with meta-analysis is the small sample size of the RCTs included, which limits the generalizability of the results. Another important limitation is the heterogeneity between the different RCTs in terms of the clinical characteristics of the DFUs included (depth, infection, and ischemia), study follow-up time, time of application, and frequency of application associated with the type of ultrasound used (contact or noncontact). Finally, the lack of certain information in the studies is another limitation in evaluating some variables since it prevents the inclusion of some studies in this meta-analysis. Further clinical trials that have a low risk of bias and use of control groups, with clear randomization and blinding of results, are needed to

determine the clinical relevance of these findings. (The study by Rastogi et al., 2019, which was previously cited in this policy, is included in this systematic review and meta-analysis.)

Messa et al. (2019) conducted a retrospective case series in patients undergoing lower extremity wound treatment with direct, low-frequency (22.5 kHz), high-intensity (approximately 60 W/cm²), ultrasonic debridement. The study included 82 wounds in 51 patients. The primary outcomes included wound healing, which was assessed by the percentage of wound reduction up to 180 days post ultrasound treatment; readmission rates; and reoperation rates. The average wound age at initial presentation was 1,013 days, with an average wound size of 9.0 cm × 7.4 cm. At 180 days post debridement, 60% of wounds had completely healed. Readmission for wound healing (70%) was primarily for further debridement (41%). Wound infection (30%) was the most common readmission for wound complications (30%). Reoperations primarily consisted of treatments for further wound healing (96%). The authors concluded that the use of direct, low-frequency, high-intensity, ultrasonic debridement is a safe and reliable adjunctive therapy for the management of wounds. The study had a small sample size and no comparison group.

Bekara et al. (2017) conducted a systematic review to describe, evaluate, and compare three recently developed methods for the management of chronic wounds: hydrosurgery (Versajet), ultrasound therapy (MIST Therapy System), and plasma-mediated bipolar radiofrequency ablation therapy (Coblation). In January 2016, an electronic database search was conducted using MEDLINE, PubMed Central, and Embase for articles concerning these three innovative methods for the management of chronic wounds. In total, 389 references were identified, and 15 articles were included. The authors extracted data regarding the number and age of individuals, indications, operating time, number of procedures, costs, wound healing time, decrease in exudation, perioperative blood loss, bacterial load, and the occurrence of complications. The 15 articles included studies that involved 563 individuals who underwent hydrosurgery (seven studies), ultrasound therapy (six studies), or Coblation (two studies). Six RCTs were included that compared the use of a scalpel or curette with hydrosurgery (two studies) or ultrasound therapy (six studies). The authors concluded that hydrosurgery, in addition to being a very precise and selective tool, allowed significantly faster debridement. Ultrasound therapy provided a significant reduction in exudation and improved the wound healing time. However, no comparative study dedicated to Coblation was identified. The authors concluded that despite the obvious clinical interest in this topic, their review of the current literature revealed a lack of prospective randomized studies that compared these devices with each other or with standard techniques, particularly for Coblation and hydrosurgery.

Cullum and Liu (2017) performed a systematic review to investigate whether ultrasound helps to heal or improve the symptoms of venous leg ulcers. Eleven RCTs, involving a total of 969 individuals, were found. Eight studies compared ultrasound with the use of no ultrasound for venous leg ulcers, and the other three compared ultrasound with sham ultrasound. Seven of the 11 studies were at a high risk of bias, one study was at a low risk of bias, and bias could not be determined in three studies due to poor reporting. The authors found that the results of one study (337 individuals) suggest that high-frequency ultrasound may be associated with more adverse events such as pain and skin redness (moderate-quality evidence). The two studies that evaluated low-frequency ultrasound did not report whether individuals experienced side effects. It is also uncertain whether either high- or low-frequency ultrasound affects individuals' quality of life. The outcomes of adverse effects, quality of life, and cost were not reported for low-frequency ultrasound treatment. The authors concluded that it is uncertain whether therapeutic ultrasound (either high or low frequency) improves the healing of venous leg ulcers. They rated most of the evidence as low or very low quality due to a risk of bias and imprecision. Most of the studies did not have many individuals, had short follow-up times, and had a poor study design.

Alkahtani et al. (2017) conducted a systematic review to examine ultrasound for the treatment of chronic wounds, therapeutic efficacies, and clinical considerations. Overall, 40 studies were included. The authors found that the evidence for the effectiveness of ultrasound for pressure ulcers is limited. RCTs showed no significant differences between the treatment groups. Low-frequency ultrasound techniques have been used in combination with standard wound care medications for the treatment of purulent wounds. The findings of these studies support the ultrasound technique as an adjunctive or alternative treatment for purulent wounds due to its therapeutic effectiveness. Studies on ultrasound for the treatment of diabetic wounds had a lack of strong evidence due to an insufficient sample size, a short follow-up period, nonrandom allocation to treatment arms, a nonblinded outcomes evaluation, a poor description of control, and concurrent interventions. The review found that the therapeutic efficacy of the low-frequency, low-dose ultrasound technique in chronic venous leg ulcers can shorten the healing period of open wounds. The authors concluded that although early studies have been relatively promising, the main challenge for developing ultrasound-based techniques as standard treatment options for different wounds is defining an exact dose response for each wound. Further controlled trials that include larger sample sizes are necessary.

Chang et al. (2017) performed a systematic review of the current clinical evidence on the use of low-frequency (20-60 kHz) ultrasound in chronic wounds. Overall, 25 studies were included. The authors found that the current body of evidence supports the use of low-frequency ultrasound as adjunctive therapy at least three times a week in the treatment

of chronic wounds. The majority (21 of 25 studies) of the evidence is limited by study design, representing mostly level 3 to level 5 evidence. One of the major limitations of the study is its inability to discern the efficacy of treatment on different types of wound etiology due to the lack of sufficient study numbers for the pooling of data. More well-designed clinical trials are needed.

Wagner-Cox et al. (2017) performed a retrospective study to examine the effect of noncontact low-frequency ultrasound (NLFU) on deep tissue pressure injury (DTPI), both hospital acquired and those present on admission (POA). Medical records from 44 adult patients with a DTPI treated with NLFU were reviewed; 22 had a hospital-acquired DTPI (HADTPI), and 22 had DTPI POA. Data were collected from the medical records, including demographic and relevant clinical characteristics, DTPI measurements, and DTPI evolution/resolution. All patients with HADTPI and DTPI POA treated with NLFU had a statistically significant decrease in injury size from initiation to discontinuation of NLFU therapy (24.6 cm vs 14.4 cm). No statistically significant difference in wound resolution was found between HADTPI vs DTPI POA (27% vs 18%). The mean size of both HADTPI and DTPI POA decreased significantly from 15.9 to 13.4 cm by NLFU therapy. Wounds were classified as resolved at completion of treatment in 23% (10 of 44) of all treated patients. Of all patients with the potential to be resolved (not discharged early or died), 63% (10 of 16) had wounds that were classified as resolved. The authors concluded that the study findings suggest that NLFU is a viable and promising treatment option for both HADTPI and DTPI POA. The findings are limited by the lack of a control group receiving a different treatment. Future studies are needed to confirm these results and to examine the efficacy and feasibility of DTPI across care settings.

White et al. (2016) compared NLFU in addition to SOC three times a week, with SOC alone at least once weekly in a single-site, blinded RCT. Overall, 36 randomized participants with chronic venous ulcers completed treatment (17 NLFU plus SOC; 19 SOC). NLFU plus SOC participants experienced a -47% change in wound area, and those receiving SOC experienced a -39% change, with a difference of -7.4% ($p = 0.6$). The median number of infections per participant was two in both groups, and the change in quality-of-life scores was not significant. NLFU plus SOC participants had a substantial mean reduction in pain score of -14.4 points, and SOC participants' pain scores reduced by -5.3, with a difference of -9.1 ($p = 0.08$). The authors concluded that the results demonstrated the importance of high-quality wound care and that outcome measures favored NLFU plus SOC over SOC; however, the differences were not statistically significant. The significance of this study is limited by a small sample size and short follow-up period.

Beheshti et al. (2014) conducted a study in 90 participants with venous leg ulcers who were randomized to receive standard treatment (consisting of compression therapy) and high-frequency ultrasound; standard treatment and MIST ultrasound therapy; or standard treatment alone. The study end points included the mean duration of wound healing, edema, pain, size of ulcers, and recurrence rate of the ulcers. In the two ultrasound groups, therapy was administered three times per week until the wound healed. Wound size, pain, and edema were assessed at baseline and after 2 and 4 months. The mean duration of complete wound healing was 8.13 months in the standard treatment group, 6.10 months in the high-frequency ultrasound group, and 5.70 months in the MIST ultrasound group. Edema was mild to severe in all groups at the first visit following treatment. After 4 months, the edema was less in both ultrasound groups compared with the standard treatment group; however, the difference of edema between the two ultrasound groups was not significant. Pain degree was also found to be decreased in the ultrasound groups compared with the standard treatment group, but again, no significant differences were found between the two ultrasound groups. Six months following treatment, the venous leg ulcers recurred in four participants in the standard treatment group and two participants in each of the ultrasound groups. Although the authors noted improvement in edema, a decrease in pain, and less recurrence in the ultrasound groups compared with those who received compression therapy, there were no significant differences between the high-frequency ultrasound and the MIST ultrasound groups. The findings are limited by the lack of masking to group assignment. Further studies that include longer follow-up are needed.

Yao et al. (2014) conducted a randomized clinical pilot study designed to determine the effects of NLFU devices when used for the treatment of chronic, nonhealing wounds. Participants were randomly assigned to one of three groups: application of NLFU three times per week (group 1), NLFU once per week (group 2), and the control (group 3) that received no NLFU. All participants received standard wound care plus off-loading for a total of 4 weeks. Percent area reduction (PAR) of each wound compared with baseline was evaluated weekly. Twelve participants with DFUs, two (16.7%) with type 1 and 10 (83.3%) with type 2 diabetes, with an average age of 58 ± 10 years and a total of 12 foot ulcers, were enrolled. Group 1 had significant WAR at weeks 3, 4, and 5 compared with baseline, with the greatest PAR being 86% ($p < 0.05$); groups 2 and 3 had 25% PAR and 39% PAR, respectively, but there were no statistically significant differences between group 2 and group 3 over time. Based on the information provided by this small, randomized study, NLFU is effective in treating neuropathic DFUs through at least, in part, inhibition of proinflammatory cytokines in chronic wounds and improvement of tissue regeneration. Therapeutic application of NLFU three times per week allows for the best WAR. The results of this study must be confirmed in a larger trial.

Watson et al. (2011) assessed the clinical effectiveness of weekly delivery of low-dose, high-frequency therapeutic ultrasound in conjunction with standard care for hard-to-heal venous leg ulcers in a multicenter, two-arm RCT. The study included 337 participants with at least one venous leg ulcer of > 6 months' duration or > 5-cm² area and an ankle-brachial pressure index of ≥ 0.8 . The study evaluated weekly administration of low-dose, high-frequency ultrasound therapy for up to 12 weeks plus standard care compared with standard care alone. The two groups had no significant difference in the time to healing of the reference leg ulcer. After adjustment for baseline ulcer area, baseline ulcer duration, use of compression bandaging, and study center, there was still no evidence of a difference in time to healing. There was no difference in time to complete healing of all ulcers, with a median time to healing of 328 days with standard care and 365 days with ultrasound. There was no evidence of a difference in rates of recurrence of healed ulcers (17 of 31 with ultrasound vs 14 of 31 with standard care). There was also no difference between the two groups in health-related quality of life, both for the physical component score and the mental component score, but there were significantly more adverse events in the ultrasound group. The authors concluded that low-dose, high-frequency ultrasound administered weekly for 12 weeks during dressing changes, in addition to standard care, did not increase ulcer healing rates, affect quality of life, or reduce ulcer recurrence.

Noncontact Real-Time Fluorescence Wound Imaging

The safety and efficacy of handheld, noncontact imaging devices that can visualize fluorescent bacteria and measure wound surface area in real time have not been established in the published literature. All fluorescence wound imaging devices, regardless of U.S. Food and Drug Administration approval, require further clinical studies for evaluation to ensure that their safety and efficacy are validated. While some evidence exists for the predictive characteristics of the method compared with conventional wound cultures, the clinical utility of the method in improving care and individuals' outcomes is unclear. In 2025 (updated 2026), a Hayes Health Technology Assessment found that the overall body of clinical evidence for the MolecuLight autofluorescence imaging (AFI) devices for detection of pathogenic bacteria in chronic lower extremity wounds is of very low quality and provides insufficient support to draw firm conclusions regarding clinical validity or clinical utility. Across multiple cross-sectional studies, MolecuLight demonstrated generally low and highly variable diagnostic accuracy, although several studies suggest that it may be more effective than reliance on clinical signs and symptoms (CSS) alone for identifying wounds with high bacterial loads. Evidence for improved outcomes in individuals was limited to two poor- or very poor-quality clinical utility studies, including one small, open-label RCT and one retrospective cohort study, which suggest possible increases in wound healing rates and reductions in amputations but did not report clear or statistically significant differences between groups. Safety concerns were minimal, as the device is noncontact, although false-negative results may delay appropriate care. No definitive selection criteria for individuals were identified. Overall, Hayes concluded that while MolecuLight shows potential clinical value as an adjunctive assessment tool, more robust, well-designed studies are needed to establish its accuracy, impact on outcomes, and appropriate role in wound management.

Badrie et al. (2025) conducted a systematic review to evaluate the clinical utility and diagnostic accuracy of AFI for detecting bacterial burden in acute and chronic wounds. Using PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)-guided methodology, the authors reviewed 17 quantitative studies published between 2015 and 2023, including clinical trials, observational studies, and case reports, with most evaluating the MolecuLight i:X device and one assessing the PRODIGI system. Across diverse wound types (e.g., DFUs, venous leg ulcers, burns), AFI consistently demonstrated higher sensitivity and accuracy than white light inspection and CSS alone for identifying moderate to heavy bacterial loads [$\geq 10^4$ colony-forming units (CFUs)/g], particularly in wounds without overt clinical signs of infection. Several studies showed that fluorescence-guided swabbing improved microbiological yield; AFI-guided debridement resulted in significant reductions in bacterial burden; and AFI-informed treatment decisions were associated with improved wound healing outcomes. Quality appraisal using Evidence-Based Librarianship criteria found 10 studies to be methodologically valid, although heterogeneity in study design, sample sizes, diagnostic thresholds, and microbiological reference standards precluded meta-analysis. The authors concluded that AFI is a promising adjunctive tool that enhances wound assessment, supports targeted clinical interventions, and may improve infection management but emphasized the need for larger, independent, multicenter RCTs to better define its impact on long-term healing and infection control outcomes. Key limitations include small sample sizes, lack of standardized fluorescence interpretation, potential publication and industry funding bias, and limited data on long-term outcomes and cost-effectiveness.

Oropallo et al. (2025) conducted a prospective, single-center cohort study aimed to evaluate the association between high bacterial colonization and wound-associated pain in venous ulceration using MolecuLight i:X for bacterial detection. The authors evaluated 46 adults with venous ulceration of the lower extremity who self-reported a wound-associated pain score of ≥ 4 on a scale of 1 to 10. Before any treatments were performed (e.g., debridement), participants rated their pain during the study visit, and fluorescence images were captured. Regions of pain and positive fluorescence signals were sketched onto a printed wound image. Fluorescence imaging was repeated post procedure, and participants rerated their pain either at the end of the study visit or over the phone the following day. Semiquantitative analysis involved visual estimation of the percentage overlap between regions of fluorescence and pain in the wound bed. Wilcoxon matched-

pairs signed rank tests and Mann-Whitney t tests assessed changes in pain scores post procedure. Fluorescence from elevated bacterial loads and biofilm was present in every venous ulcer assessed, usually covering $\leq 50\%$ of the wound bed and commonly colonizing the wound edges. Regions of pain were more extensive than regions of fluorescence in the wound bed, and some degree of overlap was identified in 40 of 46 participants (87%). This overlap was often substantial (29 participants with $> 25\%$ overlap and 16 with $> 50\%$ overlap). Overall mean pain scores were 8.17 before the procedure and 6.87 after the procedure, corresponding to a 1.30-point reduction that was statistically significant ($p < 0.0001$). Pain score reduction was higher when participants rerated their pain 1 day after debridement (3.40-point reduction; $p = 0.004$). In conclusion, the authors observed that fluorescence signals from clinically significant bacterial colonization and biofilms were commonly present in painful venous lower extremity ulcerations. Regions of participant-reported pain and positive fluorescence frequently overlapped, suggesting a relationship between the two. Wound-associated pain scores were immediately reduced after objectively targeted bacterial removal via real-time fluorescence imaging, with an even greater reduction observed by the next day. The authors concluded that the MolecuLight i:X imaging device offers a practical method to enhance the management of venous ulcers through its noninvasive, real-time fluorescent visualization. Understanding the association between chronic bacterial presence and pain in venous ulcers can inform treatment and management strategies, potentially enhancing individuals' quality of life and satisfaction, promoting healing, and reducing complications. This study has limitations. First, the relatively small sample size may limit the generalization of the results. Data collection was limited to treatment during a single appointment, which does not account for the full scope of wound management. Longitudinal studies should be conducted to establish the role of imaging-informed treatment in the context of consistent, multidisciplinary management of chronic venous ulcers.

Hanson-Viana et al. (2024) conducted a single-center, prospective observational study to investigate the use of a real-time fluorescence imaging device, MolecuLight i:X, as a predictive tool for skin graft integration through detection of common pathogens associated with burn wound infection and graft failure. This study included adult participants with burns and previously infected wounds that were deemed clinically and microbiologically clean; therefore, these participants were candidates for grafting. Prior to grafting, a fluorescence imaging assessment (blinded to the surgical team) localized areas that were positive for moderate-high bacterial loads ($> 10^4$ CFUs/g). The most common pathogens found in the initial infection were *Pseudomonas aeruginosa* (40%), *Escherichia coli* (16%), and *Enterobacter cloacae* (13%). Intraoperatively, a standard swab sample from the recipient site was collected by the surgical team. Post operation, areas that were positive/negative for fluorescence and areas of graft take and failure were overlapped and measured (cm^2) over a two-dimensional schematic. The performance and accuracy of fluorescence imaging and swab sampling in relation to graft outcomes were assessed. Overall, 38 participants were enrolled in this study, with 73% ($n = 28$) being male; the mean (SD) age was 42 ± 13 years. The most common burn causes were direct fire contact [73% ($n = 28$)], electrical [18% ($n = 7$)], and contact burns [8% ($n = 3$)]. Six participants had comorbidities: three participants with psychiatric disorders (schizophrenia, depressive disorder, or substance withdrawal syndrome), one with hypertension, and three with type 2 diabetes. The mean total body surface area involvement was $14.5\% \pm 12.4\%$ (range, 0.8%-40.2%). Overall, 25 of the 38 participants enrolled had complete graft take, while 13 had partial graft losses. There were no total losses. Fluorescence imaging was positive in 100% of losses vs 31% (four of 13) of the swab microbiology. Fluorescence imaging was found to have a sensitivity of 86%, specificity of 98%, positive predictive value of 72%, negative predictive value of 99%, and an accuracy of 94% for predicting any type or range of graft loss in the entire cohort. Meanwhile, the sensitivity of microbiology from swab samples was 30%, with a specificity of 76%. The authors concluded that real-time fluorescence imaging of bacterial and biofilm presence and location can identify areas, with high sensitivity and specificity, of the wound bed in which grafts are most likely to fail or succeed. Better diagnostic methods that flag preinfection states and enable proactive bacterial and biofilm management may lead to better outcomes and support the viability of early excision. However, further studies are needed to prove the association of fluorescence imaging with other burn wound outcomes, including research that compares fluorescence imaging with other advanced microbe typification methods or that investigates variations in graft take between beds prepared using fluorescence guidance and those following standard care protocols.

Armstrong et al. (2023; included in the Hayes report cited above) conducted a post hoc, multicenter clinical trial analysis of 138 DFUs to evaluate the role of fluorescence imaging in detecting biofilm-encased and planktonic bacteria in wounds at high loads. The sensitivity and specificity of clinical assessment and fluorescence imaging were compared across bacterial loads of concern (10^4 - 10^9 CFUs/g). Quantitative tissue culture confirmed the total loads. Bacterial presence was confirmed in 131 of 138 ulcers. Of these, 93.9% had loads of $> 10^4$ CFUs/g. In those wounds, symptoms of infection were largely absent and did not correlate with or increase proportionately with bacterial loads at any threshold. Fluorescence imaging increased sensitivity for the detection of bacteria across loads of 10^4 to 10^9 CFUs/g ($p < 0.0001$), peaking at 92.6% for loads of $> 10^8$ CFUs/g. Imaging further showed that 84.2% of ulcers contained high loads in the periwound region. The authors anticipate that the definition of chronic inhibitory bacterial load will spark a paradigm shift in DFU wound assessment and management that encourages and enables earlier intervention along the bacterial-infection continuum; therefore, sequelae of infection may be prevented, and DFU outcomes may be improved. The authors concluded that fluorescence imaging of bacterial burden has the potential for facilitating early bacterial intervention,

monitoring treatment effectiveness during and after debridement, aiding antimicrobial stewardship to limit antibiotic and antimicrobial dressing prescriptions, and improving wound healing outcomes. Clinicians had limited experience using fluorescence imaging in a clinical context before the study; this may have lowered the sensitivity of fluorescence imaging to detect bacteria at loads of $> 10^4$ CFUs/g (sensitivity previously reported to range from 72%-100%). Limitations of the imaging technology described include a limited (1.5 mm) depth of excitation and the inability to detect non-porphyrin-producing bacteria, including all species from the *Streptococcus*, *Enterococcus*, and *Finnegoldia* genera, although these rarely occur monomicrobially in chronic wounds. Additional limitations are that this study focused primarily on high bacterial load as a contributor to wound pathogenicity, even though there are additional systemic factors that delay DFU healing and increase infection risk (e.g., peripheral artery disease, poor glycemic control, neuropathy). As the number of data points for each bacterial load threshold ranges from 14 to 34, these results should be interpreted with caution. The clinical utility of the technology to improve patient-centered outcomes was not assessed in this study. Finally, there is a risk of bias and potential conflict of interest, as this clinical trial was funded by MolecuLight, Inc.

Chen et al. (2023) conducted a meta-analysis to assess the effect of ultrasound-supported wound debridement (USSD) in individuals with DFUs. A comprehensive literature examination through January 2023 was implemented, and 1,873 linked studies were appraised. The chosen studies contained 577 individuals with DFUs in the studies' baseline; 282 of them were receiving USSD, 204 were receiving standard care, and 91 were receiving a placebo. ORs, in addition to 95% CIs, were used to calculate the consequence of USSD in individuals with DFUs by the dichotomous styles and a fixed- or random-effects model. The USSD applied to DFUs caused a significantly higher wound healing rate compared with the standard care (OR, 3.08; 95% CI, 1.94-4.88; $p < 0.001$), with no heterogeneity ($I^2 = 0\%$), and the placebo (OR, 7.61; 95% CI, 3.11-18.63; $p = 0.02$), also with no heterogeneity ($I^2 = 0\%$). The USSD applied to DFUs caused a significantly higher wound healing rate than the standard care and the placebo. Precautions should be taken, and the consequences should be considered, as all the picked studies for this meta-analysis had small sample sizes. A limitation of this meta-analysis is potential selection bias because a number of the studies involved in the meta-analysis were not covered. In addition, bias may have been increased due to the inclusion of missing or erroneous data from prior studies. The findings of this study need to be validated by well-designed studies. Further investigation is needed before the clinical usefulness of this procedure is proven. (Rastogi et al., 2019, and Yao et al., 2014, previously cited in this policy, are included in this meta-analysis.)

Derwin et al. (2023) conducted a prospective observational study to investigate WAR outcomes in a complex wound population composed of nonhealing acute and chronic wounds. The relationship between bacterial autofluorescence signals and WAR was investigated. Area measurements were collected both manually and digitally, and both methods were compared for accuracy. Overall, 26 participants with 27 wounds of varying etiologies were observed twice weekly for 2 weeks. Digital wound measurement, wound bacterial status assessment, and targeted debridement were performed through a point-of-care fluorescence imaging device (MolecuLight i:X). The WAR rate was calculated using baseline and last-visit measurements. Statistical analyses, including t tests, Fisher exact tests, Wilcoxon signed rank test for method comparison, and analysis of variance for bacterial subgroups, were applied as pertinent. The overall average WAR was -3.80 cm² or a decrease of 46.88% (manual measurement) and -2.62 cm² or a 46.05% decrease (digital measurement via MolecuLight). There were no statistically significant differences between the WAR of acute and chronic wounds ($p = 0.7877$). A stepwise correlation between the WAR and bacterial status classification per fluorescence findings was observed, in which persistent bacteria resulted in worse WAR outcomes. An overestimation of wound area by manual measurement was 23% on average. The authors concluded that fluorescence imaging signals were linked to WAR outcome and could be considered predictive. Wounds exhibiting bacterial loads that persisted at the end of the study period had worse WAR outcomes, while those with bacterial loads effectively removed with management demonstrated greater WAR. Manual measurement of the wound area consistently overestimated wound size compared with digital measurement. However, if the measurement was performed by the same operator, the overestimation was uniform enough that the WAR was calculated to be close to accurate. Notwithstanding, single wound measurements are likely to result in overestimation. Limitations to this study include its single-site design and a small, heterogeneous sample in relation to the age and wound etiology. Therefore, caution is needed in generalizing the results. Further research that includes RCTs is needed to validate these findings.

Ramirez-GarciaLuna et al. (2023) conducted a multicenter prospective study in 66 outpatient wound care participants using hyperspectral imaging to collect visible light, thermography, and bacterial fluorescence images. Wounds were assessed and screened using the International Wound Infection Institute checklist for CSS of infection. Principal component analysis was performed on the images to identify wounds presenting as infected, inflamed, or noninfected. The model could accurately predict all three wound classes (infected, inflamed, and noninfected) with an accuracy of 74%. They performed best on infected wounds (sensitivity, 100%; specificity, 91%) compared with noninflamed (sensitivity, 94%; specificity, 70%) and inflamed wounds (sensitivity, 85%; specificity, 77%). The authors concluded that combining multiple imaging modalities enables the application of models to improve wound assessment. Infection detection by CSS is vulnerable to subjective interpretation and variability based on clinicians' education and skills.

Enabling clinicians to use point-of-care hyperspectral imaging may allow earlier infection detection and intervention, possibly preventing delays in wound healing and minimizing adverse events. Limitations to the research include a lack of systematic, objective infection measurements, such as tissue biopsies, as the classification of infected vs noninfected wounds was clinically done. The findings of this study need to be validated by well-designed studies. Further investigation is needed before the clinical usefulness of this procedure is proven.

Ai-Jalodi et al. (2021) conducted a multicenter, prospective pilot study that evaluated the time to healing over 12 weeks in participants with long-standing DFUs. The aim of this study was to assess the efficacy and safety of a porcine peritoneum-derived matrix in DFU treatment. In addition to weekly assessments for wound size, investigators analyzed bacterial burden using the MolecuLight i:X wound imaging device and bacterial protease testing. Participants received a weekly application of Meso Wound Matrix Scaffold, a lyophilized, porcine peritoneum-derived matrix (DSM Biomedical Inc.), for up to 8 weeks. Descriptive statistics were chosen for this analysis. Overall, 12 male participants and three female participants, with an average age of 57 years, were enrolled during a 2-month period. The average wound duration was 30 weeks. Due to unrelated health issues, four participants were withdrawn. For the study end point of complete wound closure at 12 weeks, six (55%) of the remaining 11 participants achieved complete closure, and four participants (36%) healed during the 8-week treatment period. The average number of cellular- and/or tissue-based product (CTP) applications was six. Participants who healed all had negative bacterial protease by 9 weeks and no fluorescence on MolecuLight i:X, indicating a low bacterial load. The authors concluded that this small pilot study indicates that individuals with long-standing DFUs may respond to a porcine peritoneal-derived CTP. In this study, the CTP appears to have inhibited bacterial growth in the wound; however, further research is needed. The limitations of this pilot study include the small sample size, lack of a control arm, and loss of four of the 15 participants to unrelated adverse events. The bacterial results require further study before drawing any conclusions. Further research that includes RCTs is needed to validate these findings.

In 2021 (updated 2024), ECRI's Clinical Evidence Assessment suggested that the evidence for the use of the MolecuLight i:X Fluorescence Imaging System is inconclusive. Studies provide insufficient evidence to determine improvement in individuals' outcomes. While the evidence might suggest that the MolecuLight i:X Fluorescence Imaging System may be helpful for identification of wounds with bacterial loads, additional RCTs are needed to confirm the safety and efficacy of the device. Le et al. (2020; included in ECRI report above) conducted a prospective, multicenter, observational study on the use of MolecuLight i:X in 350 individuals. Wounds underwent CSS assessment using the International Wound Infection Institute checklist, immediately followed by fluorescent imaging with the MolecuLight device. CSS assessment missed approximately 85% of bacterial loads that were greater than 10^4 CFUs/g, which can be indicative of infection. The authors found that the use of the MolecuLight device resulted in higher sensitivity and accuracy for the detection of the bacteria. Limitations of the study include the underreporting of bacteria diversity with the culture analysis, limited experience of clinicians in using the MolecuLight device, and funding of the study by MolecuLight, Inc. The authors recommended that the MolecuLight device be used in combination with CSS assessment and that evidence from larger, longitudinal studies would be beneficial. The clinical utility of the technology to improve patient-centered outcomes was not assessed in this study.

Farhan and Jeffery (2020) conducted a single-center observational study to assess the MolecuLight i:X device for efficacy in pediatric burn wounds and the overall feasibility of the device. Ten individuals were recruited, and the device was used in 16 different wounds to assess for the presence or absence of CSS of infection; swabs were obtained to confirm the findings. The authors found that the device demonstrated the ability to visually identify significant bacterial growth and high adherence for use of the device. The authors concluded that these findings may pave the way for the use of bacterial fluorescence imaging in the pediatric burn population. However, the clinical utility of the technology to improve patient-centered outcomes was not assessed in this study.

Chew et al. (2020) stated that early diagnosis of wound infections is crucial, as wound infections have been shown to increase individuals' morbidity and mortality. These researchers examined the use of MolecuLight i:X to identify infections in acute open wounds in hand trauma. In a prospective cohort study, data were collected from participants who visited the hand trauma unit over a 4-week period before having surgery. Wounds were inspected for clinical signs of infection, and autofluorescence images were taken using the MolecuLight i:X device. Wound swabs were taken, and the results were interpreted according to a report by a microbiologist. Autofluorescence images were interpreted by a clinician blinded to the microbiology results. Overall, 31 participants were included, and data, which were based on 35 wounds, were collected; three wounds (8.6%) showed positive clinical signs of infection, three (8.6%) were positive on AFI, and two (5.7%) of the wound swab samples were positive for significant infection. AFI correlated with clinical signs and wound swab results for 34 wounds (97.1%). In one case, the clinical assessment and AFI showed positive signs of infection, but the wound swabs were negative. The authors concluded that AFI in acute open wounds may be useful to provide real-time confirmation of bacterial infection and thus guide management.

Pijpe et al. (2019) performed a pilot study that compared the detection of bacteria in burn wounds between MolecuLight i:X, which is a bacterial fluorescence imaging device, and standard microbiological swabs. A total of 14 individuals, with 20 wounds, participated in the study. Wounds were swabbed three times: once with a standard swab, once with a high-fluorescent area swab, and, finally, with a nonfluorescent area swab. The proportion agreement of the microbiological results was calculated, and the accuracy of the device to detect relevant bacteria was assessed. The diagnostic accuracy of the bacterial fluorescence imaging device to detect relevant bacteria in burn wounds was moderate, and the reliability was equal to that of standard swabbing. Further research in larger sample sizes is needed to establish the safety and efficacy of the fluorescence imaging device.

In a pilot study, Serena et al. (2019) evaluated 19 wounds for the diagnostic accuracy of wound bacteria when bacterial fluorescence imaging (MolecuLight i:X) was used in combination with evaluation of CSS. CSS criteria for wounds to determine the presence or absence of moderate to heavy bacterial loads were determined using the NERDS (nonhealing, exudate, red and bleeding surface or granulation tissue, debris, and smell) and STONEES (size, temperature, osteomyelitis, new areas, exudate, erythema, and smell) methods. Then, fluorescence images of the wound were acquired, along with determination of bacterial presence or absence. Biopsies were obtained under local anesthetic and sent to the laboratory for confirmation; all laboratory staff was blinded to the wound's assessment outcomes. Four of the 19 participants (21%) were identified as positive (for moderate to heavy bacterial loads) based on CSS alone. The use of fluorescence imaging in combination with CSS assessment led to 2.5- to 3.2-fold improvements in reported diagnostic accuracy measures compared with CSS assessment alone. The authors concluded that the data in this pilot study suggest that current SOC assessment for wounds fails to identify many wounds with moderate to heavy bacterial loads, leaving individuals with undetected and untreated bacteria. The addition of bacterial fluorescence imaging improved the sensitivity and accuracy of assessments for detecting moderate to heavy bacterial loads. Limitations of this study include the small sample size, which was not statistically significant, and lack of follow-up. Future studies that include larger samples are needed.

In a prospective observational study, Hurley et al. (2019) swabbed 43 wounds in 33 participants. The authors wanted to establish the accuracy of the wound imaging device at detecting bacteria. All data were collected in the outpatient wound care clinic setting. Participants aged over 18 years, with a variety of wounds, were recruited; participants on antibiotics for wound infection were excluded. Images from the wounds were captured with the handheld fluorescent device; on visualization of bacteria, areas of red or cyan fluorescence that indicated bacteria were swabbed and sent to the laboratory for culture and sensitivity testing. Of the swabs taken, 95.4% were positive for bacteria growth, and nine different species of bacteria were identified. Limitations include device incompatibility for wounds with active bleeding, dressings that contained silver (a potent antimicrobial), and the sample size. Despite these limitations, the authors concluded that the device is safe, effective, and accurate for use. Further research should be directed to its application in other environments such as preoperative and perioperative settings.

For information on current clinical trials studying the use of MolecuLight i:X and bacterial growth, go to [ClinicalTrials.gov](https://clinicaltrials.gov).

Clinical Practice Guidelines

Association for the Advancement of Wound Care (AAWC)

Members of the AAWC, Wound Healing Society, and Canadian Association for Enterostomal Therapy formed the International Consolidated Guidelines Taskforce in 2015 to update the *AAWC Venous Ulcer Guideline* to the collaborative, intersociety, endorsed *International Consolidated Venous Ulcer Guideline*. This clinical practice guideline contains systematically developed recommendations intended to optimize patient care and assist physicians and other health care practitioners and participants to make decisions about appropriate health care for venous ulcer clinical care.

The recommendations include the following:

- Low-frequency ultrasound may support healing, reduce pain, and improve quality of life in those with nonhealing venous or mixed-etiology venous ulcers.
- Use ultrasound stimulation in combination with adequate patient-appropriate compression and moisture-retentive dressings to add possible venous ulcer healing benefit, but be aware that limited evidence supports cost-effectiveness, enduring benefit, or parameters of application.
- Warming therapy has insufficient evidence of healing efficacy to inform venous ulcer management decisions about its use as an adjunct to optimal patient-appropriate compression and moisture-retentive dressings.

(AAWC, 2015; Couch et al., 2017)

The AAWC also indicates that ultrasound stimulation may be used in combination with adequate patient-appropriate compression and moisture-retentive dressings to add possible venous ulcer healing benefit, but be aware that limited evidence supports enduring benefit or parameters of application (AAWC, 2015).

In 2010, the AAWC released a guideline on the care of pressure ulcers. While noncontact ultrasound therapy was included as a potential second-line treatment if first-line treatments failed to induce wound healing, the strength of the evidence supporting this decision was low (level C), indicating that there is limited evidence for this technology.

American Podiatric Medical Association (APMA)

The APMA, in collaboration with the Society for Vascular Surgery and Society for Vascular Medicine, developed a 2016 clinical practice guideline on the management of the diabetic foot. Their recommendations do not include warming therapy, NNWT, or low-frequency ultrasound for the treatment of wounds (Hingorani et al., 2016).

National Institute for Health and Care Excellence (NICE)

A NICE Medtech Innovation Briefing (MIB212) concluded that there is limited evidence to show whether MolecuLight i:X reduces wound closure time or improves antibiotic stewardship (2020).

The NICE guideline for diabetic foot problems prevention and management recommends one or more of the following as standard care for treating DFUs: off-loading; control of foot infection; control of ischemia; wound debridement; and wound dressings. NICE recommends that negative pressure wound therapy should be considered after surgical debridement for DFUs, on the advice of the multidisciplinary foot care service. Dermal or skin substitutes as an adjunct to standard care can be considered when treating DFUs, only when healing has not progressed, and on the advice of the multidisciplinary foot care service (NICE, 2019; reaffirmed July 3, 2025).

In NICE guidance on the MIST Therapy System for the promotion of wound healing, NICE states that the MIST Therapy System shows potential to enhance the healing of chronic, hard-to-heal, complex wounds compared with standard methods of wound management. However, the amount and quality of published evidence on the relative effectiveness of the MIST Therapy System are not sufficient, at this time, to support the case for routine adoption of the MIST Therapy System (NICE, 2011; reaffirmed June 17, 2016).

U.S. Food and Drug Administration (FDA)

This section is to be used for informational purposes only. FDA approval alone is not a basis for coverage.

Warm-Up Active Wound Therapy (Augustine Medical, Inc.) received 510(k) approval from the FDA on March 28, 1997, as a wound and burn occlusive heated dressing. Refer to the following website for more information (use product code MSA): <http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfPMN/pmn.cfm>. (Accessed March 11, 2026)

The MIST Therapy System is regulated by the FDA as a Class II device and is classified as an ultrasound wound cleaner. This device was approved via the FDA 510(k) process in April 2004. In May 2005, the FDA granted marketing clearance to Celleration for the MIST Therapy System 5.1, with an expanded indication. The approved indication for use is to produce "a low-energy ultrasound-generated mist to promote wound healing through wound cleansing and maintenance debridement by the removal of yellow slough, fibrin, tissue exudates, and bacteria." In 2014, the FDA approved the UltraMIST® System (K140782), which has a smaller, sleeker, and user-friendly design. Refer to the following website for more information: http://www.accessdata.fda.gov/cdrh_docs/pdf5/K050129.pdf. (Accessed March 11, 2026)

Refer to the following website for additional devices (use product code NRB): <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfPMN/pmn.cfm>. (Accessed March 11, 2026)

The FDA cleared the MolecuLight i:X device under its 510(k) premarket notification process as substantially equivalent to predicate devices. For additional information, refer to the following: https://www.accessdata.fda.gov/cdrh_docs/pdf19/K191371.pdf. (Accessed March 11, 2026)

Refer to the following website for additional devices (use product code QJF or FXN): <http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfPMN/pmn.cfm>. (Accessed March 11, 2026)

References

Ai-Jalodi O, Sabo M, Patel K, et al. Efficacy and safety of a porcine peritoneum-derived matrix in diabetic foot ulcer treatment: a pilot study. *J Wound Care*. 2021 Feb;30(Sup2):S18-S23.

Alkahtani S, Kunwar PS, Jalilifar M, et al. Ultrasound-based techniques as alternative treatments for chronic wounds: a comprehensive review of clinical applications. *Cureus*. 2017 Dec;9(12):e1952.

Armstrong DG, Edmonds ME, Serena TE. Point-of-care fluorescence imaging reveals extent of bacterial load in diabetic foot ulcers. *Int Wound J*. 2023 Feb;20(2):554-566.

Association for the Advancement of Wound Care (AAWC). Pressure ulcer guideline. October 2010.

Association for the Advancement of Wound Care (AAWC). Venous ulcer guideline. December 2015.

Badrie S, Moore Z, Patton D, et al. The clinical utility of autofluorescence imaging for bacterial detection in wounds: a systematic review. *Int Wound J*. 2025 Jun;22(6):e70678.

Beheshti A, Shafigh Y, Parsa H, et al. Comparison of high-frequency and MIST ultrasound therapy for the healing of venous leg ulcers. *Adv Clin Exp Med*. 2014 Nov-Dec;23(6):969-975.

Bekara F, Vitse J, Fluieraru S, et al. New techniques for wound management: a systematic review of their role in the management of chronic wounds. *Arch Plast Surg*. 2018 Mar;45(2):102-110.

Chang YR, Perry J, Cross K. Low-frequency ultrasound debridement in chronic wound healing: a systematic review of current evidence. *Plast Surg (Oakv)*. 2017 Feb;25(1):21-26.

Chen H, Yu Z, Liu N, et al. The efficacy of low-frequency ultrasound as an added treatment for chronic wounds: a meta-analysis. *Int Wound J*. 2023 Feb;20(2):448-457.

Chew BJW, Griffin M, Butler PE, et al. The use of MolecuLight i:X device in acute hand trauma. *J Plast Reconstr Aesthet Surg*. 2020 Jul;73(7):1357-1404.

Couch K, Corbett L, Gould L, et al. The International Consolidated Venous Ulcer Guideline update 2015: process improvement, evidence analysis, and future goals. *Ostomy Wound Manage*. 2017 May;63(5):42-46.

Cullum n and Liu Z. Therapeutic ultrasound for venous leg ulcers. *Cochrane Database Syst Rev*. 2017 May;5(5):CD001180.

Derwin R, Patton D, Strapp H, et al. Integrating point-of-care bacterial fluorescence imaging-guided care with continued wound measurement for enhanced wound area reduction monitoring. *Diagnostics (Basel)*. 2023 Dec;14(1):2.

ECRI. Clinical Evidence Assessment. MolecuLight i:X Fluorescence Imaging System (MolecuLight, Inc.) for managing chronic wounds. February 2021.

Ellis SL, Finn P, Noone M, et al. Eradication of methicillin-resistant *Staphylococcus aureus* from pressure sores using warming therapy. *Surg Infect (Larchmt)*. 2003 Spring;4(1):53-55.

Farhan n and Jeffery S. Utility of MolecuLight i:X for managing bacterial burden in pediatric burns. *J Burn Care Res*. 2020 Feb;41(2):328-338.

Flores-Escobar S, Álvaro-Afonso FJ, García-Álvarez Y, et al. Ultrasound-assisted wound (UAW) debridement in the treatment of diabetic foot ulcer: a systematic review and meta-analysis. *J Clin Med*. 2022 Mar;11(7):1911.

Gibbons GW, Orgill DP, Serena TE, et al. A prospective, randomized, controlled trial comparing the effects of noncontact, low-frequency ultrasound to standard care in healing venous leg ulcers. *Ostomy Wound Manage*. 2015 Jan;61(1):16-29.

Haesler E. Local warming/heat therapy for chronic wounds: a WHAM evidence summary. *Wound Pract Res*. 2023;31(2):90-92.

Hanson-Viana E, Rojas-Ortiz JA, Rendón-Medina MA, et al. Bacterial fluorescence imaging as a predictor of skin graft integration in burn wounds. *Burns*. 2024 Sep;50(7):1799-1811.

Hayes, Inc. Clinical Research Response. MolecuLight i:X Imaging Device (MolecuLight) for bacterial imaging. December 2020. Archived January 30, 2022.

Hayes, Inc. Health Technology Assessment. MolecuLight (MolecuLight Corp.) for assessment of pathogenic bacteria in chronic lower extremity wounds. November 2025; updated April 2026.

Hingorani A, LaMuraglia GM, Henke P, et al. The management of diabetic foot: a clinical practice guideline by the Society for Vascular Surgery in collaboration with the American Podiatric Medical Association and the Society for Vascular Medicine. *J Vasc Surg*. 2016 Feb;63 (2 Suppl):3S-21S.

Hurley CM, McClusky P, Sugrue RM, et al. Efficacy of a bacterial fluorescence imaging device in an outpatient wound care clinic: a pilot study. *J Wound Care*. 2019 Jul;28(7):438-443.

Karr JC. External thermoregulation of wounds associated with lower-extremity osteomyelitis: a pilot study. *J Am Podiatr Med Assoc*. 2003 Jan-Feb;93(1):18-22.

Le L, Baer M, Briggs P, et al. Diagnostic accuracy of point-of-care fluorescence imaging for the detection of bacterial burden in wounds: results from the 350-patient fluorescence imaging assessment and guidance trial. *Adv Wound Care (New Rochelle)*. 2021 Mar;10(3):123-136.

Messa CA, Chatman BC, Rhemtulla IA, et al. Ultrasonic debridement management of lower extremity wounds: retrospective analysis of clinical outcomes and cost. J Wound Care. 2019 May;28(Sup5):S30-S40.

MolecuLight website. <https://moleculight.com>. Accessed March 11, 2026.

Murphy CA, Houghton P, Brandys T, et al. The effect of 22.5 kHz low-frequency contact ultrasound debridement (LFCUD) on lower extremity wound healing for a vascular surgery population: a randomized controlled trial. Int Wound J. 2018 Jun;15(3):460-472.

National Institute for Health and Care Excellence (NICE). MolecuLight i:X for wound imaging, Medtech Innovation Briefing (MIB212). Published June 18, 2020.

National Institute for Health and Care Excellence (NICE). NG19. Diabetic foot problems: prevention and management. January 2016. Updated January 18, 2023.

National Institute for Health and Care Excellence (NICE). The MIST Therapy System for the promotion of wound healing. London, UK: NICE. July 2011. Reaffirmed June 2016.

Olyaie M, Rad FS, Elahifar MA, et al. High-frequency and noncontact low frequency ultrasound therapy for venous leg ulcer treatment: a randomized, controlled study. Ostomy Wound Manage. 2013 Aug;59(8):14-20.

Oropallo AR, Lee PJ, Rao A, et al. Unveiling the relationship between pain and bacterial load in venous ulcers with implications in targeted treatment. J Vasc Surg Venous Lymphat Disord. 2025 Feb;13(4):102213.

Pijpe A, Ozdemir Y, Sinnige JC, et al. Detection of bacteria in burn wounds with a novel handheld autofluorescence wound imaging device: a pilot study. J Wound Care. 2019 Aug;28(8):548-554.

Ramirez-GarciaLuna JL, Martinez-Jimenez MA, Fraser RDJ, et al. Is my wound infected? A study on the use of hyperspectral imaging to assess wound infection. Front Med (Lausanne). 2023 Aug;10:1165281.

Rastogi A, Bhansali A, Ramachandran S. Efficacy and safety of low-frequency, noncontact airborne ultrasound therapy (Glybetac) for neuropathic diabetic foot ulcers: a randomized, double-blind, sham-control study. Int J Low Extrem Wounds. 2019 Mar;18(1):81-88.

Serena TE, Harrell K, Serena L, et al. Real-time bacterial fluorescence imaging accurately identifies wounds with moderate-to-heavy bacterial burden. J Wound Care. 2019 Jun;28(6):346-357.

Thomas DR, Diebold MR, Eggemeyer LM. A controlled, randomized, comparative study of a radiant heat bandage on the healing of stage 3-4 pressure ulcers: a pilot study. J Am Med Dir Assoc. 2005 Jan-Feb;6(1):46-49.

Wagner-Cox P, Duhamel HM, Jamison CR, et al. Use of noncontact low-frequency ultrasound in deep tissue pressure injury: a retrospective analysis. J Wound Ostomy Continence Nurs. 2017 Jul/Aug;44(4):336-342.

Watson JM, Kang'ombe AR, Soares MO, et al. Use of weekly, low dose, high frequency ultrasound for hard to heal venous leg ulcers: the VenUS III randomized controlled trial. BMJ. 2011 Mar;342:d1092.

White J, Ivins N, Wilkes A, et al. Non-contact low-frequency ultrasound therapy compared with UK standard of care for venous leg ulcers: a single-center, assessor-blinded, randomized controlled trial. Int Wound J. 2016 Oct;13(5):833-842.

Yao M, Hasturk H, Kantarci A, et al. A pilot study evaluating noncontact low frequency ultrasound and underlying molecular mechanism on diabetic foot ulcers. Int Wound J. 2014 Dec;11(6):586-593.

Yue J, Zhang S, Sun Q, et al. Local warming therapy for treating chronic wounds: a systematic review. Medicine (Baltimore). 2018 Mar;97(12):e9931.

Policy History/Revision Information

Date	Summary of Changes
08/01/2026	Supporting Information - Updated <i>Clinical Evidence</i> and <i>References</i> sections to reflect the most current information - Archived previous policy version 2026T0521Z

Instructions for Use

This Medical Policy provides assistance in interpreting UnitedHealthcare standard benefit plans. When deciding coverage, the member specific benefit plan document must be referenced as the terms of the member specific benefit plan may differ from the standard plan. In the event of a conflict, the member specific benefit plan document governs. Before using this policy, check the member specific benefit plan document and any applicable federal or state mandates.

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